

Lower Bounds for Gradient Descent

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- Yuhan Ye (MIT)



(2026). “Silver Rate Is (Almost) Optimal for Gradient Descent: The Strongly Convex Case”. In: *arXiv preprint arXiv:2609.26873*.



(2026a). “Improved Gradient Descent Lower Bounds Beyond Nesterov”. In: *arXiv preprint arXiv:2609.02855*.



(2026b). “Silver Rate Is (Almost) Optimal for Gradient Descent”. In: *arXiv preprint arXiv:2609.09152*.

Gradient Descent with Predetermined Stepsizes

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex L -smooth function. Consider the problem

$$\min_{x \in \mathbb{R}^d} f(x).$$

¹WLOG we assume $L = 1$ by replacing f with f/L and each stepsize h_t with Lh_t .

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Choose $h_1, \dots, h_n \geq 0$ before seeing f , then run

$$x_t = x_{t-1} - h_t \nabla f(x_{t-1}), \quad t = 1, \dots, n.$$

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Let $\mathcal{F}_L(\mathbb{R}^d)$ be the class of convex L -smooth functions on $\mathbb{R}^d$ with a nonempty set of minimizers¹. The **worst-case performance** for the schedule $H = (h_1, \dots, h_n)$ is quantified by

$$\mathcal{R}_n(H) := \sup_{d \in \mathbb{N}} \sup_{f \in \mathcal{F}_L(\mathbb{R}^d)} \sup_{x_* \in \arg \min f} \sup_{x_0 \in \mathbb{R}^d \setminus \{x_*\}} \frac{f(x_n) - f(x_*)}{\frac{L}{2} \|x_0 - x_*\|^2}. \quad (1)$$

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$\Omega(1/n)$ Lower Bound for Constant Stepsizes

Set $S = \sum_{t=1}^n h_t$ and $\delta = (1 + 2S)^{-1}$. Consider the convex 1-smooth Huber function, with $x_0 = 1$, $x_* = 0$:

$$f(x) = \begin{cases} x^2/2, & |x| \leq \delta, \\ \delta|x| - \delta^2/2, & |x| > \delta. \end{cases}$$

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Every iterate stays in the linear tail, where $\nabla f(x) = \delta$:

$$x_t = 1 - \delta \sum_{j=1}^t h_j \geq 1 - \delta S = \frac{1 + \delta}{2} \geq \delta.$$

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Theorem (Drori and Teboulle 2014)

$$\mathcal{R}_n(H) \geq 2f(x_n) = 2\delta(1 - \delta S) - \delta^2 = \delta = \frac{1}{1 + 2S}.$$

As a consequence, if all $h_t \leq 1$, then $\mathcal{R}_n(H) \geq (2n + 1)^{-1}$.

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In fact, if $h_t = 1$, then one can show that $R_n(H) = (2n + 1)^{-1}$ (Drori and Teboulle 2014).

$\Omega(1/n^2)$ Lower Bound for General First-Order Methods

Set $d = 2n + 1$, $x_0 = 0$. Consider the convex 1-smooth quadratic chain

$$f(x) = \frac{1}{8} \left[(1 - x^{(1)})^2 + \sum_{i=1}^{d-1} (x^{(i)} - x^{(i+1)})^2 + (x^{(d)})^2 \right],$$

with minimizer $x_*^{(i)} = 1 - i/(d+1)$ and $\|x_*\|^2 = \frac{d(2d+1)}{6(d+1)} \leq \frac{2(n+1)}{3}$.

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Key idea: Each iteration reveals at most one new coordinate (Nemirovsky and Yudin 1983)

$$E_k = \text{span}\{e_1, \dots, e_k\}, \quad \nabla f(E_k) \subset E_{k+1} \implies x_k \in E_k.$$

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On E_k , the $k+1$ differences sum to 1; equal differences minimize their squared sum. Hence

$$\min_{x \in E_k} f(x) = \frac{1}{8(k+1)}, \quad f(x_n) - f(x_*) \geq \frac{1}{8(n+1)} - \frac{1}{8(d+1)} = \frac{1}{16(n+1)}.$$

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Theorem (Nesterov 2004)

$$\mathcal{R}_n(H) \geq \frac{2[f(x_n) - f(x_*)]}{\|x_*\|^2} \geq \frac{3}{16(n+1)^2}.$$

The Silver Stepsize Schedule

Let $\rho = 1 + \sqrt{2}$. Build the schedule recursively:

$$H_1 = (\sqrt{2}), \quad H_{k+1} = (H_k, 1 + \rho^{k-1}, H_k), \quad k \geq 1.$$

The schedule H_k has $n = 2^k - 1$ iterations.

Theorem (Altschuler and Parrilo 2025b)

$$\mathcal{R}_n(H_k) \leq \frac{2}{1 + \sqrt{4\rho^{2k} - 3}} \leq \frac{1}{(n+1)^{p_{\text{sil}}}}, \quad p_{\text{sil}} = \log_2 \rho \approx 1.27155.$$

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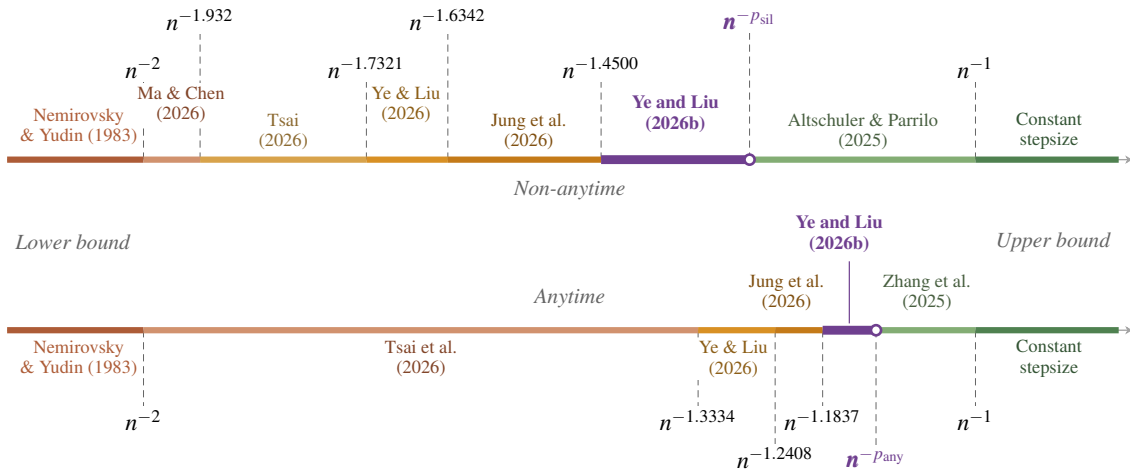
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Altschuler and Parrilo conjectured that p_{sil} is the optimal polynomial exponent.

Smooth Convex Lower Bounds



Smooth Convex Lower Bounds

Theorem (Ye and Liu 2026b)

For every nonnegative n -step schedule H ,

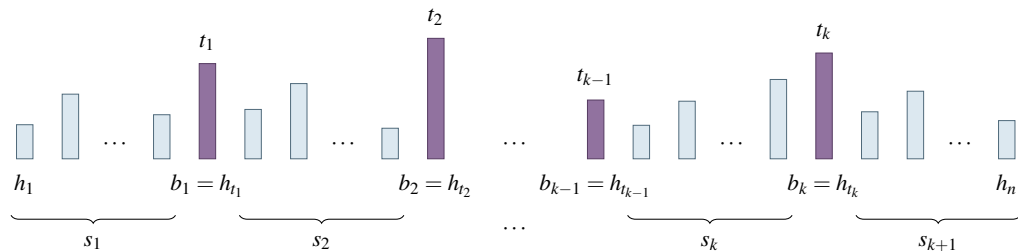
$$\mathcal{R}_n(H) \geq n^{-p_{\text{sil}} - o(1)}.$$

For any infinite schedule, at infinitely many horizons,

$$\mathcal{R}_n(H_n) \geq n^{-p_{\text{any}} - o(1)}, \quad p_{\text{any}} = \frac{2p_{\text{sil}}}{1 + p_{\text{sil}}} \approx 1.1195.$$

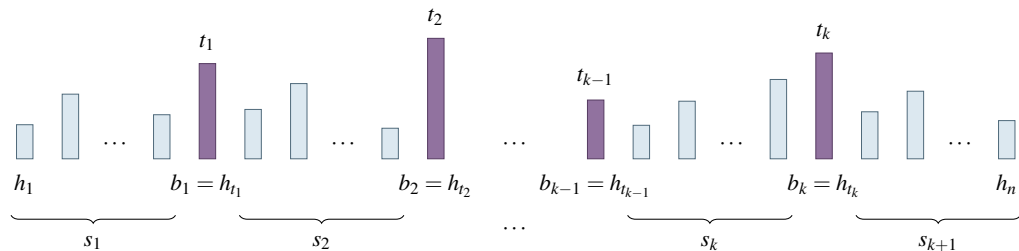
The Checkpoint Method (Ma and Chen 2026)

Given any schedule, select checkpoints at $T = \{t_1 < \dots < t_k\}$.



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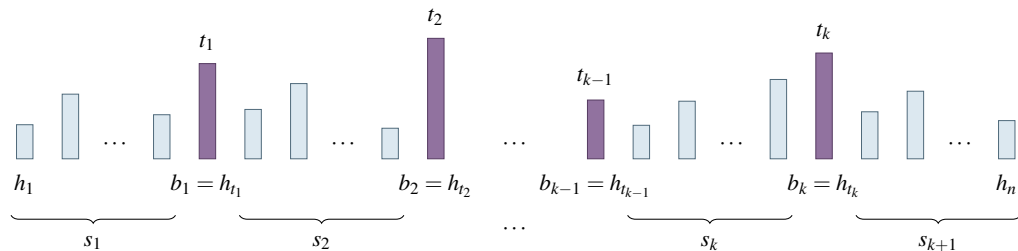
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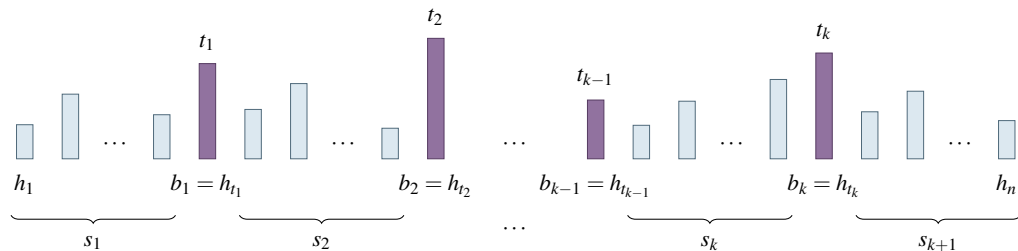
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- Forcing GD to explore a new dimension at the selected checkpoints T gives a tight lower bound.
- Maximizing over all possible sets of checkpoints gives a tight lower bound.

The Hard Function: A Chain of Components

Similar to Nesterov's chain, our hard function sums smooth convex components on neighboring pairs:

$$F(x) = \frac{1}{2} \sum_{i=1}^k \Phi_i(x^{(i)}, x^{(i+1)}) + \frac{1}{2} \underbrace{H_\delta(x^{(k+1)} - \ell_{k+1})}_{\text{terminal term}}.$$

Here $H_\delta(z) = 0$ for $z \leq 0$; for $z > 0$, it agrees with the previous Huber function.

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Insist $\Phi_i \geq 0$ and $\min F = 0$. The terminal term gives

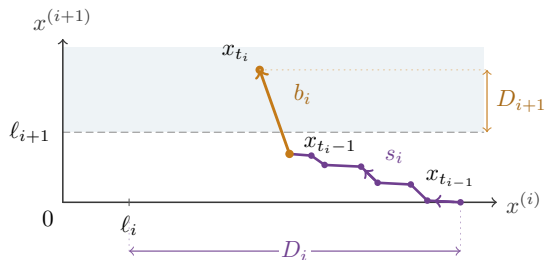
$$F(x_n) - \min F \geq \frac{1}{2} H_\delta(\underbrace{x_n^{(k+1)} - \ell_{k+1}}_{D_{k+1}}).$$

A larger D_{k+1} provides a better lower bound.

From One Component to the Next

During the i -th gap

Φ_i moves $x^{(i)}$ downward and $x^{(i+1)}$ upward; later components are inactive.

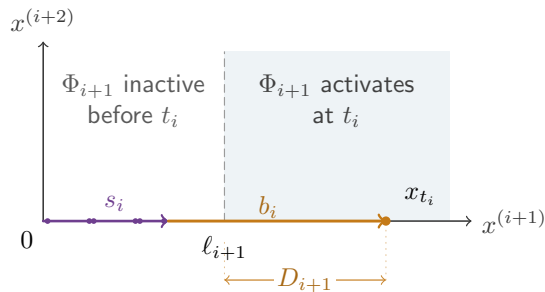


At the checkpoint t_i

The selected step b_i raises $x^{(i+1)}$ enough to activate Φ_{i+1} .

In the next gap

Φ_{i+1} takes over $x^{(i+1)}$, $x^{(i+2)}$.



(With $x_0 = e_1$, set $\ell_1 = 0$ and $D_1 = 1$; subsequent ℓ_i, D_i can be defined inductively.)

One-Sided Huber Construction (Jung, Cho, and Yun 2026)

Let $X = x^{(i)}$, $Y = x^{(i+1)}$ and $\Phi_i(X, Y) = \frac{1}{2}H_{\delta_i}(X - Y - \ell_i)$. In the affine region,

$$\frac{1}{2}\nabla\Phi_i = \frac{\delta_i}{4}(1, -1).$$

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The gap starts at $(X, Y) = (\ell_i + D_i, 0)$. During the gap, Y rises by $\delta_i s_i/4$. Earlier components can only push X upward, so the margin $X - Y - \ell_i$ drops by at most $\delta_i s_i/2$.

Keep the component affine by choosing

$$D_i - \frac{\delta_i s_i}{2} = \delta_i \quad \implies \quad \delta_i = \frac{2D_i}{s_i + 2}.$$

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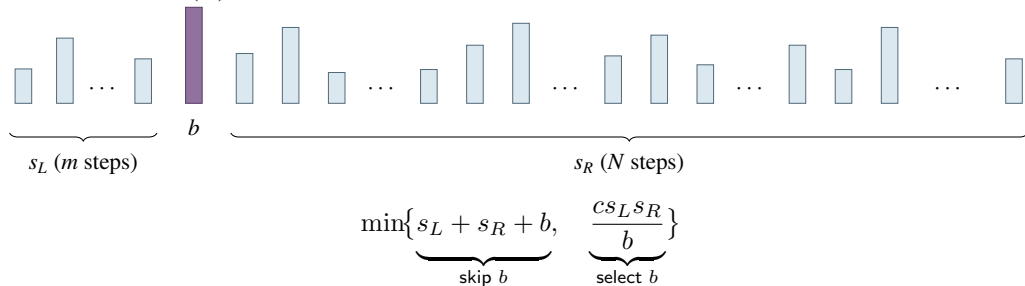
$$D_i - \frac{\delta_i s_i}{2} = \delta_i \implies \delta_i = \frac{2D_i}{s_i + 2}.$$

Set the next activation threshold $\ell_{i+1} = Y_{t_i-1} = \delta_i s_i / 4$. The output excess becomes:

$$D_{i+1} = \frac{\delta_i b_i}{4}, \quad \boxed{\frac{D_{i+1}}{D_i} = \frac{b_i}{\textcolor{red}{c}(s_i + 2)}}, \quad \textcolor{red}{c} = 2.$$

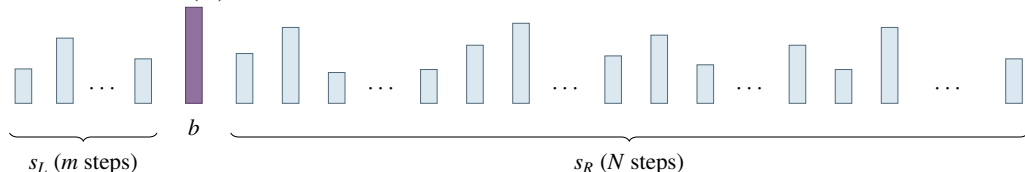
The Critical Schedule Is Balanced

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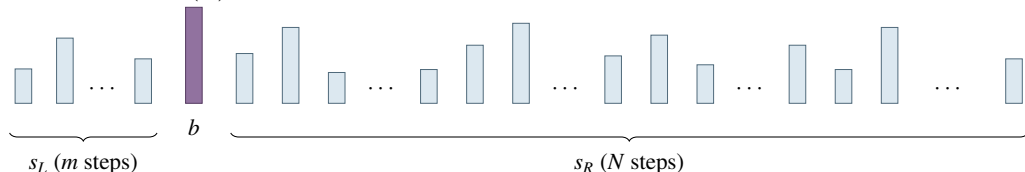
$$\min \left\{ \underbrace{s_L + s_R + b}_{\text{skip } b}, \underbrace{\frac{cs_L s_R}{b}}_{\text{select } b} \right\}$$

If $m \ll N$, then $s_L \approx m^p \ll s_R \approx N^p$. The critical schedule makes the two choices tie:

$$b \approx cs_L, \quad s_L + s_R + b \approx N^p + (1 + c)m^p.$$

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The first $m + 1$ iterations add only $\sim m^p$ to N^p . The power law would instead allow

$$(m + N + 1)^p - N^p \approx pN^{p-1}(m + 1) \gg m^p.$$

Therefore, a critical schedule cannot have a very uneven split.

The Silver Exponent

Consider two identical m -step blocks, each with stepsize sum s :

$$H_{2m+1} = (H_m, b, H_m), \quad \underbrace{2s + b}_{\text{skip } b} \approx \underbrace{\frac{cs^2}{b}}_{\text{select } b}.$$

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Let $\rho_c = (2s + b)/s$. The relation above gives

$$\rho_c(\rho_c - 2) = c \quad \implies \quad \rho_c = 1 + \sqrt{1 + c}.$$

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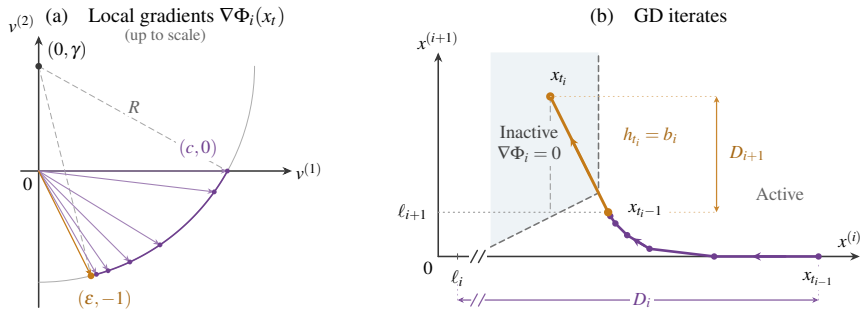
$$\rho_c(\rho_c - 2) = c \implies \rho_c = 1 + \sqrt{1 + c}.$$

The stepsize sum grows by ρ_c when the number of steps roughly doubles. Its reciprocal predicts the exponent

$$\rho_c^{-k} = (2^k)^{-\log_2 \rho_c} \approx n^{-p(c)}, \quad p(c) = \log_2(1 + \sqrt{1 + c}).$$

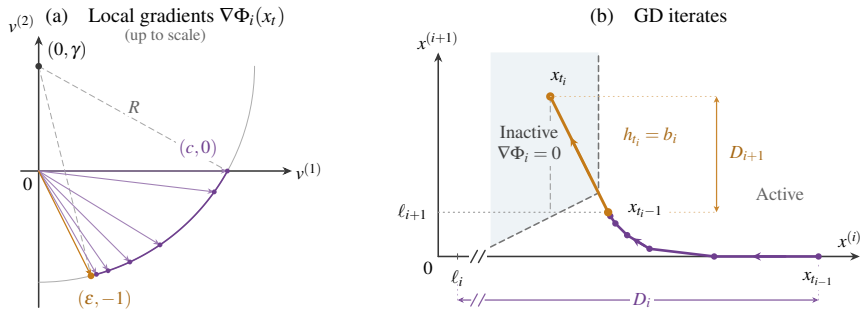
Crucially, $c = 1$ gives the silver exponent.

Bending the Trajectory (Ye and Liu 2026b)



The gradient starts (near) horizontal and approaches $(\epsilon, -1)$ just before the checkpoint.

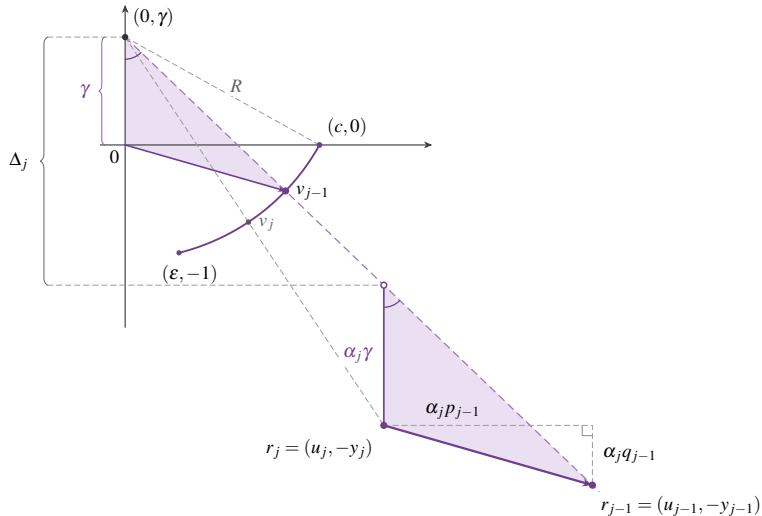
Bending the Trajectory (Ye and Liu 2026b)



The gradient starts (near) horizontal and approaches $(\epsilon, -1)$ just before the checkpoint. Most of the checkpoint step is utilized to increase the output. Consequently,

$$c \text{ improves from } 2 \text{ to } \sqrt{1 + 2\gamma + \epsilon^2} \rightarrow 1.$$

Construction of a Reference Trajectory



$$u_{j-1} = u_j + \alpha_j p_{j-1}, \quad y_{j-1} = y_j + \alpha_j q_{j-1}$$

Build backward from the pre-checkpoint gradient $(\epsilon, -1)$.

The shaded triangles scale by the stepsize α_j . The preceding gradient stays on the arc.

A Smooth Function for the Reference Trajectory

For a convex function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, define its **Moreau envelope** which is convex and 1-smooth

$$f(z) = \min_w \left\{ g(w) + \frac{1}{2} \|z - w\|^2 \right\}.$$

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For the **support function** $g(z) = \max_{v \in K} \langle v, z \rangle$ with K compact and convex,

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Set

$$\mathcal{V} = \{0, v_0, \dots, v_m, (c, 0)\}, \quad K = \text{conv } \mathcal{V}.$$

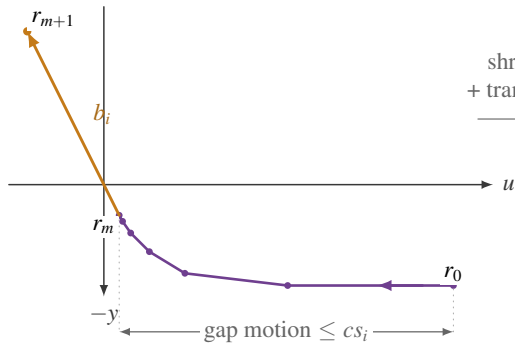
The previous construction gives

$$\Pi_K(r_j) = v_j \quad \implies \quad \nabla f(r_j) = v_j.$$

Thus f realizes the reference trajectory.

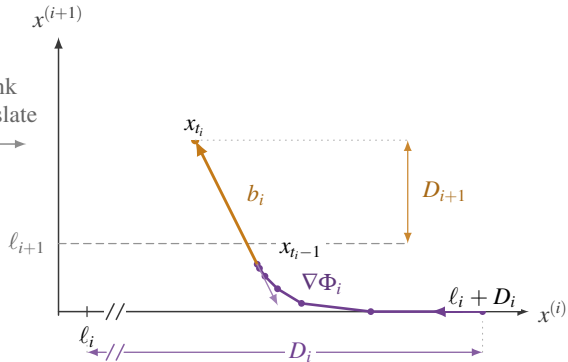
Fitting the Reference Trajectory into the Chain

Original trajectory f



shrink
+ translate
→

Scaled trajectory Φ_i



Eliminating Gradient Interference from the Previous Component

For $2 \leq i \leq k$, the previous component Φ_{i-1} also depends on the current component's input coordinate $x^{(i)}$.

$$\partial_i F(x) = \frac{1}{2} \left[\underbrace{\partial_2 \Phi_{i-1}(x^{(i-1)}, x^{(i)})}_{\text{previous component}} + \underbrace{\partial_1 \Phi_i(x^{(i)}, x^{(i+1)})}_{\text{current component}} \right].$$

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Lemma (Ye and Liu 2026b, §3.1)

$$v \geq \ell_i \implies \partial_2 \Phi_{i-1}(u, v) = 0 \quad \text{for every } u.$$

During gap i , $x_t^{(i)} \geq \ell_i$, so regardless of how earlier coordinates move, the extra gradient from the previous component is zero.

The Strongly Convex Case

Let $\mathcal{F}_{\mu,L}(\mathbb{R}^d)$ be the class of μ -strongly convex L -smooth functions on $\mathbb{R}^d$. Let $\kappa := L/\mu$ be the condition number. The worst-case performance can be quantified by

$$D_{n,\kappa}(H) := \sup_{d \in \mathbb{N}} \sup_{f \in \mathcal{F}_{1/\kappa,1}(\mathbb{R}^d)} \sup_{x_0 \neq x_*} \frac{\|x_n - x_*\|^2}{\|x_0 - x_*\|^2}, \quad (2)$$

$$E_{n,\kappa}(H) := \sup_{d \in \mathbb{N}} \sup_{f \in \mathcal{F}_{1/\kappa,1}(\mathbb{R}^d)} \sup_{x_0 \neq x_*} \frac{f(x_n) - f(x_*)}{f(x_0) - f(x_*)}. \quad (3)$$

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For a target error $0 < \varepsilon < 1$, define the iteration complexities by

$$T_D(\kappa, \varepsilon) := \inf\{n \in \mathbb{N}_{\geq 1} : \exists H \in [0, \infty)^n \text{ with } D_{n,\kappa}(H) \leq \varepsilon\},$$

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Altschuler and Parrilo 2025a established $O(\kappa^{1/p_{\text{sil}}} \log(1/\varepsilon))$ bounds for T_D and T_E .

What Strong Convexity Changes

Add a quadratic to the smooth convex chain:

$$F(x) = \frac{\|x\|^2}{2\kappa} + \frac{1 - 1/\kappa}{2} \sum_i \Phi_i(x^{(i)}, x^{(i+1)}).$$

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If the next component activates above threshold ℓ_i and the incoming excess is D_i , decay can erase that excess before the checkpoint. A large ratio ℓ_i/D_i makes this especially dangerous.

Controlling the Contraction by Blocking

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- In a block, select checkpoints as in the smooth convex case and use bending.
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- In a block, select checkpoints as in the smooth convex case and use bending.
- Regular m -step blocks lose only a constant fraction of excess.
- If no checkpoint is selected, the unselected mass is carried forward, causing the gap mass to keep growing.

Repairing a Long Unselected Gap

If carried gap mass becomes too large, insert checkpoints to restore a controlled threshold-to-excess ratio $\ell_{i+1}/D_{i+1} \approx s_i/b_i$.

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Lemma (Liu and Ye 2026, Lemma C.1)

Let $h_1, \dots, h_r \geq 0$ and $S = \sum_{t=1}^r h_t > 0$. Either some stepsize satisfies $h_t \geq 3S/4$, or there are $u < v$ with

$$h_u, h_v \geq \frac{S}{8r^2}, \quad \sum_{t=u+1}^{v-1} h_t \leq \min\{h_u, h_v\}.$$

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- Using a Huber component, the repair construction retains $\gtrsim (r+1)^{-4}$ of its excess.
- There is ≤ 1 such repair per block, causing an $O(\log m)$ log-excess loss per block.

Bridging and the Exponential Lower Bound

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$$\min\{D_{n,\kappa}(H), E_{n,\kappa}(H)\} \geq \exp\left[-C \frac{n}{m} \log(m+1)\right].$$

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- To guarantee error $\leq \epsilon$, we need $C(n/m) \log(m+1) \geq \log(1/\epsilon)$:

$$n \gtrsim \frac{m}{\log(m+1)} \log(1/\epsilon) = \kappa^{1/p_{\text{sil}} - o(1)} \log(1/\epsilon).$$

Concluding Remarks

- The arguments can be adapted to negative stepsizes.
- Can the $o(1)$ term be eliminated?

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